

# Entanglement, Decoherence, and the Thermodynamic Cost of Classical Records

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<https://github.com/hmahaffeyges/IAM-Validation>

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## Abstract

Entangled systems show correlations that violate Bell inequalities (Bell, 1964; Aspect et al., 1982; Hensen et al., 2015), ruling out local hidden-variable models. Quantum mechanics predicts these correlations exactly. What it leaves open is the quantum-to-classical transition: where and at what cost a superposition becomes a definite classical record. Decoherence theory (Zurek, 1981; Joos & Zeh, 1985; Zurek, 2003) explains the suppression of interference; Landauer’s principle (Landauer, 1961) sets the minimum thermodynamic cost of an irreversible record,  $k_B T \ln 2$  per bit. The Informational Actualization Model (IAM) applies that cost at cosmological scale: at the Gibbons–Hawking temperature  $T_H = \hbar H / 2\pi k_B$  the informational entropy produced by structure formation enters horizon thermodynamics, giving  $\mu(a) < 1$  for matter and  $\Sigma(a) = 1$  for light (Mahaffey, 2026a). Here we set out what this picture says about entanglement. It makes no change to quantum mechanics and adds no hidden variables: Bell correlations are those of the shared quantum state until an irreversible record is made. Its new content is quantitative: a gravitational decoherence time  $\tau_{\text{IAM}} = \hbar k_B^2 T^2 \ln 2 / E_G^3$  that scales as  $T^2$  and, at fixed density, as  $m^{-5}$ , against  $m^{-5/3}$  and no temperature dependence for Penrose–Diósi. For a 1 pg silica sphere at 10 mK,  $\tau_{\text{IAM}} = 559$  s and  $\tau_{\text{PD}} = 7.8$   $\mu\text{s}$ ; the two cross at  $m \approx 2.3 \times 10^{-10}$  kg. We also correct the exponent that links the cosmological activation function to structure formation: matching  $\exp(1 - 1/a)$  in matter domination requires  $n = 7/2$ .

## 1 Entanglement and the quantum-to-classical transition

Bell’s theorem (Bell, 1964) shows that no local hidden-variable theory reproduces the correlations of entangled states. Experiments confirm the violation, including a loophole-free test with electron spins 1.3 km apart (Hensen et al., 2015) and photon pairs distributed over 1,200 km by satellite (Yin et al., 2017). Quantum mechanics predicts these results without modification.

Two distinct questions are often merged. The first is why the correlations are non-classical; the answer is that an entangled pair is described by one quantum state, not by two systems with pre-existing local properties. The second is where and how that state gives way to definite classical outcomes. Decoherence explains the loss of interference through entanglement with an environment (Zurek, 1981; Joos & Zeh, 1985), and quantum Darwinism explains how redundant records make outcomes objective (Zurek, 2003, 2009). Neither fixes the thermodynamic cost of the record. Landauer’s principle does: an irreversible record costs at least  $k_B T \ln 2$  per bit, dissipated as heat (Landauer, 1961).

IAM takes this cost seriously at the largest scale. If structure formation produces classical information irreversibly, the corresponding entropy belongs in the horizon thermodynamics from which the Friedmann equations follow (Jacobson, 1995; Cai & Kim, 2005). This paper states what that does and does not imply for entanglement.

## 2 Two sectors

Gravitational decoherence in IAM requires a timelike worldline,  $d\tau > 0$ . Photons travel null geodesics,  $d\tau = 0$ , and undergo no gravitational decoherence in transit; they are absorbed and measured in the ordinary way, but propagation adds nothing to the informational entropy. Massive particles accumulate proper time and, when bound into virialised structures, contribute to it. This is the origin of the matter/photon sector split (Mahaffey, 2026c).

Local environmental decoherence (gas collisions, thermal photons, phonons) is the dominant mechanism in laboratory systems and is unchanged by IAM. IAM adds a gravitational channel and an accounting statement: every irreversible record made on a timelike worldline carries its Landauer cost, and at cosmological scale that cost appears in the horizon entropy budget.

## 3 The derivation chain

**Step 1.** Jacobson derived the Einstein equations as an equation of state from the entropy–area relation of local causal horizons (Jacobson, 1995).

**Step 2.** Cai and Kim obtained the Friedmann equations from the first law on the apparent horizon of an FRW universe (Cai & Kim, 2005).

**Step 3.** Structure formation produces classical information irreversibly. For  $1/r$  potentials the virial theorem gives  $2K = |V|$ . IAM identifies the kinetic half with the heat dissipated in the quantum-to-classical transition, priced at  $k_B T_H \ln 2$  per bit (Gibbons & Hawking, 1977).

**Step 4.** The accumulated informational contribution is described by the activation function

$$\mathcal{E}(a) = \exp\left(1 - \frac{1}{a}\right), \quad (1)$$

monotonically increasing,  $\mathcal{E}(1) = 1$ ,  $\mathcal{E}(\infty) = e$ , and negligible at early times ( $\mathcal{E}(z = 10) = 4.5 \times 10^{-5}$ ).

**Step 5.** Including this entropy in the first law adds Hubble friction to the matter perturbation equation:

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2}{H_{\Lambda\text{CDM}}^2 + \beta_m \mathcal{E}(a)} < 1, \quad \Sigma(a) = 1, \quad (2)$$

with  $\beta_m = \Omega_m/2 = 0.15765$  from the virial theorem ( $\Omega_m = 0.3153$ ). The Planck 2018 Level 2 posterior returns  $\beta_m = 0.1583 \pm 0.0033$ ,  $0.2\sigma$  from this value (Mahaffey, 2026b).

## 4 Entanglement under the two sectors

Because IAM leaves quantum mechanics unchanged, Bell correlations of a pure shared state are those of standard theory. The framework’s statement is about how long an entangled state of *massive* systems can survive the gravitational channel.

For photon pairs no gravitational decoherence acts in transit, and the CHSH parameter stays at  $|S| = 2\sqrt{2}$  for an ideal source, whatever the distance. For massive superpositions IAM predicts gravitational decoherence with the time-dependent profile of Mahaffey (2026d),

$$S(t) = 2\sqrt{2} (1 - D(t)), \quad D(t) = \mathcal{E}(t/\tau_{\text{IAM}})/e, \quad (3)$$

and timescale (Mahaffey, 2026e)

$$\tau_{\text{IAM}} = \frac{\hbar k_B^2 T^2 \ln 2}{E_G^3}, \quad E_G = \frac{Gm^2}{r}, \quad (4)$$

where  $T$  is the bath temperature. At fixed density,  $E_G \propto m^{5/3}$ , so  $\tau_{\text{IAM}} \propto T^2 m^{-5}$ ; the Penrose–Diósi time  $\tau_{\text{PD}} = \hbar/E_G \propto m^{-5/3}$  has no temperature dependence.

Table 1: Gravitational decoherence times for silica spheres ( $\rho = 2000 \text{ kg m}^{-3}$ ), computed from Eq. (4) and  $\tau_{\text{PD}} = \hbar/E_G$ .

| System                                       | $\tau_{\text{IAM}}$                                                                  | $\tau_{\text{PD}}$ | Discriminator                               |
|----------------------------------------------|--------------------------------------------------------------------------------------|--------------------|---------------------------------------------|
| 1 fg sphere ( $10^{-15} \text{ kg}$ ), 10 mK | $5.6 \times 10^{17} \text{ s}$                                                       | 0.78 s             | PD testable; IAM not                        |
| 1 pg sphere ( $10^{-12} \text{ kg}$ ), 10 mK | 559 s                                                                                | $7.8 \mu\text{s}$  | $T^2$ vs $T^0$ ; ramp vs exponential        |
| 1 pg sphere, 20 mK                           | 2,238 s                                                                              | $7.8 \mu\text{s}$  | doubling $T$ quadruples $\tau_{\text{IAM}}$ |
| Crossover, 10 mK                             | $\tau_{\text{IAM}} = \tau_{\text{PD}}$ at $m \approx 2.3 \times 10^{-10} \text{ kg}$ |                    |                                             |

The discriminating experiment is therefore a mesoscopic superposition near  $10^{-12} \text{ kg}$ : Penrose–Diósi predicts loss of coherence in microseconds, IAM predicts none on that timescale, and a temperature scan separates the two. Distance records do not discriminate: matter-qubit entanglement over 1.3 km (Hensen et al., 2015) and photon entanglement over 1,200 km (Yin et al., 2017) are both consistent with standard decoherence and photon-mediated distribution.

Entanglement of top-quark pairs at the LHC (CMS Collaboration, 2024) is consistent with IAM: the top lifetime,  $\sim 5 \times 10^{-25} \text{ s}$ , is far shorter than any gravitational decoherence time. Bell violations from path indistinguishability without conventional entanglement (Scientific Advances, 2025) concern the structure of the state and are outside the gravitational channel. Correlations between photons that never coexisted (Megidish et al., 2013) are properties of the quantum state across time; a treatment within IAM is left open.

## 5 The exponent linking structure formation to $\mathcal{E}(a)$

Write the information production rate during matter domination as  $\dot{I} \propto \rho_m D^n H$ , with  $\rho_m \propto a^{-3}$ ,  $D \propto a$  and  $H \propto a^{-3/2}$ . With the Gibbons–Hawking temperature and the

horizon area, the entropy increment per unit  $a$  scales as  $a^{n-11/2}$ , so that

$$S_{\text{info}}(a) \propto \int a^{n-11/2} da \propto a^{n-9/2}. \quad (5)$$

Matching the  $-1/a$  dependence of  $\ln \mathcal{E}(a)$  requires

$$n - \frac{9}{2} = -1 \implies n = \frac{7}{2}. \quad (6)$$

(For  $n < 9/2$  the integral is dominated by its lower limit, so  $S_{\text{info}}$  is defined up to a constant fixed by the normalisation  $\mathcal{E}(1) = 1$ .) A bottom-up value of  $n$  from the halo mass function must reproduce  $7/2$  for the chain to close; the numerical halo-collapse calculation gives an effective  $1/a$  coefficient near unity at  $\sigma_* \approx 1.5$  (Mahaffey, 2026f). Deriving  $n$  analytically from the mass function is an open problem.

## 6 Observational status

Seventeen MCMC chains (12 MGCAMB, 5 modified CAMB) were run with the Planck 2018 likelihood (TT, TE, EE + low- $\ell$  + lensing), some combined with BAO, Pantheon+ or RSD data; final Gelman–Rubin  $R - 1$  is below 0.01 for 15 and 0.023–0.024 for the other two (Mahaffey, 2026b). Results:

- $\sigma_8 = 0.7998 \pm 0.0058$  ( $\Lambda$ CDM 0.809);
- $H_0^{(\text{photon})} = 67.16 \pm 0.47$  and  $H_0^{(\text{matter})} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ;
- $\beta_m = 0.1583 \pm 0.0033$ ,  $0.2\sigma$  from  $\Omega_m/2 = 0.15765$ ;
- $\Delta\chi^2 = +0.54$  at best fit relative to  $\Lambda$ CDM.

Planck is consistent with the IAM modification but does not by itself establish it. The prediction  $\mu_0 - 1 = -0.136$  is tested by Euclid ( $\sigma(\mu_0) \approx 0.04$ ,  $3.4\sigma$ ) and Euclid with DESI ( $5.4\sigma$ );  $\Sigma_0 = 0$  is tested by Euclid weak lensing.

## 7 IAM’s law

**IAM’s law.** Every irreversible quantum-to-classical transition on a timelike world-line produces a classical record at a thermodynamic cost of at least  $k_B T \ln 2$  per bit, with  $T$  the temperature of the system that absorbs it; at cosmological scale,  $T = T_H$  and the cost enters the horizon entropy.

Consequences: (1) quantum mechanics is unchanged, and Bell correlations are those of the shared state; (2) gravitational decoherence of massive superpositions follows Eq. (4), with a  $T^2$  signature; (3)  $\mathcal{E}(a)$  increases monotonically because each record is irreversible.

## 8 Discussion

Decoherence has been known to be irreversible since the 1980s (Zurek, 1981; Joos & Zeh, 1985); horizon entropy since Bekenstein and Hawking (Bekenstein, 1973; Hawking,

1975); the thermodynamic origin of gravity since Jacobson (Jacobson, 1995). IAM asks where the Landauer cost of cosmological structure formation goes, and answers that it enters the horizon entropy budget, producing  $\mu < 1$  and  $\Sigma = 1$ . For entanglement the framework predicts nothing beyond quantum mechanics for photons, and a specific, temperature-dependent gravitational decoherence time for massive superpositions that mesoscopic experiments can test.

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## Data Availability

Chains, scripts and figures: <https://github.com/hmahaffeyges/IAM-Validation>; Zenodo DOI [10.5281/zenodo.18702042](https://doi.org/10.5281/zenodo.18702042).

## References

- Aspect, A., Dalibard, J., & Roger, G. 1982, Phys. Rev. Lett., 49, 1804
- Bekenstein, J. D. 1973, Phys. Rev. D, 7, 2333
- Bell, J. S. 1964, Physics, 1, 195
- Cai, R.-G. & Kim, S. P. 2005, JHEP, 02, 050
- CMS Collaboration 2024, Observation of quantum entanglement in top quark pair production, Rep. Prog. Phys., 87, 117801
- Gibbons, G. W. & Hawking, S. W. 1977, Phys. Rev. D, 15, 2738
- Hawking, S. W. 1975, Commun. Math. Phys., 43, 199
- Hensen, B., et al. 2015, Nature, 526, 682
- Jacobson, T. 1995, Phys. Rev. Lett., 75, 1260
- Joos, E. & Zeh, H. D. 1985, Z. Phys. B, 59, 223
- Landauer, R. 1961, IBM J. Res. Dev., 5, 183
- Mahaffey, H. W. 2026a, Horizon Thermodynamics and Gravitational Decoherence as the Origin of  $\mu < 1$ ,  $\Sigma = 1$ , Zenodo, 10.5281/zenodo.18702042
- Mahaffey, H. W. 2026b, Dual-Sector Perturbation Cosmology: CAMB Implementation and Validation, Zenodo

- Mahaffey, H. W. 2026c, The Two Faces of Time: Coordinate Time, Proper Time, and Accumulated Decoherence, Zenodo
- Mahaffey, H. W. 2026d, The Measurement Problem Dissolved: Quantum Decoherence as Irreversible Sector Crossing in the Informational Actualization Model, Zenodo
- Mahaffey, H. W. 2026e, Gravitational Decoherence from Dual-Sector Thermodynamics: Predictions for Optomechanical Experiments and Quantum Computing Architecture, Zenodo
- Mahaffey, H. W. 2026f, Quantum Darwinism at Cosmological Scales: Gravitational Decoherence, the Cosmic Horizon, and the Emergence of Classical Structure, Zenodo
- Megidish, E., et al. 2013, Phys. Rev. Lett., 110, 210403
- Bell-inequality violation from path indistinguishability, Sci. Adv., 2025
- Yin, J., et al. 2017, Science, 356, 1140
- Zurek, W. H. 1981, Phys. Rev. D, 24, 1516
- Zurek, W. H. 2003, Rev. Mod. Phys., 75, 715
- Zurek, W. H. 2009, Nature Physics, 5, 181